

**FOURTH SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
APRIL 2021**

Mathematics

ME 4C 04—MATHEMATICAL ECONOMICS

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

ME 4C 04—MATHEMATICAL ECONOMICS

(Multiple Choice Questions for SDE Candidates)

1. The lin log model and log lin model are _____ in parameters.
(A) Non-linear. (B) Linear.
(C) Functional. (D) Dependent.
2. _____ is a growth model.
(A) A linear trend model. (B) Lin log model.
(C) Log lin model. (D) None of the above.
3. Keynes postulated _____ relationship between income and consumption.
(A) Negative. (B) Positive.
(C) Non-linear. (D) Infinite.
4. In the Keynesian linear consumption function $Y = \beta_1 + \beta_2 X$, β_1 is :
(A) Slope coefficient. (B) Intercept coefficient.
(C) Output coefficient. (D) None of the above.
5. The variable appearing on the right side of the equality sign is called :
(A) Independent variable. (B) Explanatory variable.
(C) All of the above. (D) None of the above.
6. A mathematical model assumes _____ relationship between variables.
(A) Inexact. (B) Exact.
(C) Probable. (D) None of the above.
7. Regression analysis is concerned with :
(A) Study of the dependence on one variable on the other.
(B) Predicting the average value.
(C) Predicting the population mean.
(D) All of the above.

8. Regression analysis is concerned with _____ relationship among variables.
- (A) Statistical. (B) Functional.
(C) Deterministic. (D) None of the above.
9. Correlation theory is based on the assumption of :
- (A) Randomness of variables. (B) Conditional mean.
(C) Random errors. (D) Specification.
10. The law of universal regression was first introduced by :
- (A) Irwing Fisher. (B) Laspayer.
(C) Francis Galton. (D) Pearson.
11. An expected value is the same as :
- (A) Average value. (B) Standard deviation.
(C) Dispersion. (D) None of the above.
12. The regression line or curve passes through .
- (A) Origin. (B) Vertical axis.
(C) Horizontal axis. (D) Conditional means.
13. "The descriptions be kept as simple as possible until proved inadequate" corresponds to :
- (A) Occam's razor. (B) Index numbers.
(C) Regression. (D) Correlation.
14. The most popular method of constructing sample regression function in the regression analysis is :
- (A) Method of OLS. (B) Generalised squares.
(C) Ordinary regression method. (D) None of the above.
15. Homoscedasticity means _____ for disturbances.
- (A) Equal mean. (B) Equal variance.
(C) Zero mean. (D) None of the above.

16. Economic theory makes statements that are mostly :
- (A) Quantitative. (B) Qualitative.
(C) Positive. (D) None of the above.
17. Heteroscedasticity implies :
- (A) Equal spread. (B) Unequal spread.
(C) Equal mean. (D) Equal variance.
18. The numerical value of coefficient of determination lies between :
- (A) -1 and 1. (B) 0 and 1.
(C) $-\infty$ to $+\infty$. (D) $-\infty$ to 1.
19. The rejecting of a true hypothesis is called :
- (A) Type I error. (B) Type II error.
(C) Standard error. (D) Point estimation.
20. The accepting of a false hypothesis is called :
- (A) Type I error. (B) Type II error.
(C) Standard error. (D) Point estimation.

**FOURTH SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
APRIL 2021**

Mathematics

ME 4C 04—MATHEMATICAL ECONOMICS

Time : Three Hours

Maximum : 80 Marks

Part A

*Answer **all** the twelve questions.
Each question carries 1 mark.*

1. What are the two types of econometrics ?
2. Give an example of an economic model.
3. What is the conditional expectation function or the population regression function ?
4. State Gauss-Markov Theorem.
5. What is the maximum value of the co-efficient of determination ?
6. Define correlation co-efficient.
7. What are the two branches of classical theory of statistical inference ?
8. What is the mean and variance of a standard normal variable ?
9. Define 'confidence interval'.
10. What is type I error ?
11. Give an example of a regression model that not linear in the variables.
12. What is Logarithmic Reciprocal Model ?

(12 × 1 = 12 marks)

Part B

*Answer any **six** questions in two **or** three sentences.
Each question carries 3 marks.*

13. Write a short note on Keynesian consumption function.
14. What is the difference between the population and sample regression functions ? Illustrate with an example.

Turn over

15. Given $E(u_i|X_i) = 0$, show that $E(Y_i|X_i) = \beta_2 + \beta_2 X_i$.
16. Show that the mean value of the estimated $Y = \hat{Y}_i$ is equal to the mean value of the actual Y in a sample : $\{(x_i, y_i) : i = 1, 2, \dots, n\}$.
17. What are the classical normal linear regression model assumption for μ_i .
18. Under the normality assumption find the 100 (1 - α) % confidence interval for regression co-efficient β_2
19. Write a short note on Testing of hypothesis.
20. What are the three main features of reciprocal models ?
21. Find the elasticity of the log linear regression model $\ln Y = \beta_1 + \beta_2 \ln X$.

(6 × 3 = 18 marks)

Part C

*Answer any **six** questions from the following.
Each question carries 5 marks.*

22. Write any *five* properties of coefficient of correlation.
23. Show that the least-squares estimator $\hat{\beta}_2$ is linear.
24. Show that $\hat{\sigma}^2$ is an unbiased estimator of true σ^2 .
25. What are the 5 important properties of OLS Estimators under the Normality Assumption on u_i .
26. Briefly discuss about maximum likelihood estimation of Two variable regression model.
27. Find the Confidence Intervals for Regression Co-efficients β_1 .
28. Consider the following regression model : $1/Y_i = \beta_1 + \beta_2 (1/X_i) + u_i$.
 - (a) Is this a linear regression model ? Why ? Why not ?
 - (b) How would you estimate this model ?
 - (c) What is the behavior of Y as X tends to infinity ?
 - (d) Can you give an example where such a model may be appropriate ?

29. How to measure the growth rate using the LogLin model ?
30. What is Log Linear regression model ? How to measure elasticity using this model ?

(6 × 5 = 30 marks)

Part D

Answer any **two** questions from the following.

Each question carries 10 marks.

31. Discuss various steps involved in the traditional econometric methodology.
32. Write the 10 Assumptions made in the classical linear regression model.
33. In the following table, you are given the ranks of 10 students in midterm and final examinations in mathematics. Compute Spearmans co-efficient of rank correlation and interpret it.

Midterm	1	3	7	10	9	5	4	8	2	6
Final	3	2	8	7	9	6	5	10	1	4

34. Why do we employ the normality assumption on μ_i .

(2 × 10 = 20 marks)